C. Diverse Matrix

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: constructive algorithms, greedy, math, number theory | https://codeforces.com/problemset/problem/1266/C

Let a be a matrix of size $r \times c$ containing positive integers, not necessarily distinct. Rows of the matrix are numbered from 1 to r, columns are numbered from 1 to c. We can construct an array b consisting of $r + c$ integers as follows: for each $i \in [1, r]$, let b_i be the greatest common divisor of integers in the i-th row, and for each $j \in [1, c]$ let b_{r+j} be the greatest common divisor of integers in the j-th column.

We call the matrix **diverse** if all $r + c$ numbers b_k ($k \in [1, r + c]$) are pairwise distinct.

The **magnitude** of a matrix equals to the maximum of b_k.

For example, suppose we have the following matrix:

begin*pmatrix* 2 & 9 & 7
4 & 144 & 84 end*pmatrix*

We construct the array b:

1. b_1 is the greatest common divisor of 2, 9, and 7, that is 1;
2. b_2 is the greatest common divisor of 4, 144, and 84, that is 4;
3. b_3 is the greatest common divisor of 2 and 4, that is 2;
4. b_4 is the greatest common divisor of 9 and 144, that is 9;
5. b_5 is the greatest common divisor of 7 and 84, that is 7.

So $b = [1, 4, 2, 9, 7]$. All values in this array are distinct, so the matrix is diverse. The magnitude is equal to 9.

For a given r and c, find a diverse matrix that minimises the magnitude. If there are multiple solutions, you may output any of them. If there are no solutions, output a single integer 0.

Input

The only line in the input contains two space separated integers r and c ($1 \le r,c \le 500$) — the number of rows and the number of columns of the matrix to be found.

Output

If there is no solution, output a single integer 0.

Otherwise, output r rows. The i-th of them should contain c space-separated integers, the j-th of which is $a_{i,j}$ — the positive integer in the i-th row and j-th column of a diverse matrix minimizing the magnitude.

Furthermore, it must hold that $1 \le a_{i,j} \le 10^9$. It can be shown that if a solution exists, there is also a solution with this additional constraint (still having minimum possible magnitude).

Examples

Input
2 2

Output
4 12 2 9

Input
1 1

Output
0

Note

In the first example, the GCDs of rows are $b_1 = 4$ and $b_2 = 1$, and the GCDs of columns are $b_3 = 2$ and $b_4 = 3$. All GCDs are pairwise distinct and the maximum of them is 4. Since the GCDs have to be distinct and at least 1, it is clear that there are no diverse matrices of size 2 × 2 with magnitude smaller than 4.

In the second example, no matter what $a_{1,1}$ is, $b_1 = b_2$ will always hold, so there are no diverse matrices.